````
---
title: "Machine Learning Assignment 3"
output: html_document
date: "2026-10-15"
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```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

#Import data into R.
```{r}
UniversalBank <- read.csv("Downloads/UniversalBank.csv")
```

#I an remnaming certain columns because the assignment only allows online banking, credit cards and loan acceptance.
```{r}
bank <- UniversalBank[, c("Online", "CreditCard", "Personal.Loan")]
names(bank) <- c("Online", "CC", "Loan")
head(bank)
```

#I am applying the yes/no groups.
```{r}
bank$Online <- factor(bank$Online, levels = c(0, 1))
bank$CC <- factor(bank$CC, levels = c(0, 1))
bank$Loan <- factor(bank$Loan, levels = c(0, 1))
```

#The instructions wanted 60% training and 40% validation.
```{r}
set.seed(123)
train_index <- sample(
  seq_len(nrow(bank)),
  size = 0.60 * nrow(bank)
)
train <- bank[train_index, ]
validation <- bank[-train_index, ]
nrow(train)
nrow(validation)
```

#I did customer counts with CC and Loan as rows and online as columns.
```{r}
table_A <- with(train, table(CC, Loan, Online))
ftable(
  table_A,
  row.vars = c("CC", "Loan"),
  col.vars = "Online"
)
```
#I split up the loan accepters in this group with everyone included in the group.
````

```{r}
accepted <- table_A["1", "1", "1"]
total_group <- sum(table_A["1", , "1"])
prob_B <- accepted / total_group
prob_B
```

#I am using this table to compare loan with online and another loan with CC.
```{r}
loan_online <- with(train, table(Loan, Online))
loan_online
loan_CC <- with(train, table(Loan, CC))
loan_CC
```

#I put the 6 probabilities listed in part D
```{r}
p_CC1_Loan1 <- loan_CC["1", "1"] / sum(loan_CC["1", ])
p_Online1_Loan1 <- loan_online["1", "1"] / sum(loan_online["1", ])
p_Loan1 <- mean(train$Loan == "1")
p_CC1_Loan0 <- loan_CC["0", "1"] / sum(loan_CC["0", ])
p_Online1_Loan0 <- loan_online["0", "1"] / sum(loan_online["0", ])
p_Loan0 <- mean(train$Loan == "0")
c(
  "i" = p_CC1_Loan1,
  "ii" = p_Online1_Loan1,
  "iii" = p_Loan1,
  "iv" = p_CC1_Loan0,
  "v" = p_Online1_Loan0,
  "vi" = p_Loan0
)
```

#Naive Bayes calculates probabilities for each outcomes seperately. I also divided the acceptance score by the total of both scores.
```{r}
accept_score <- p_CC1_Loan1 * p_Online1_Loan1 * p_Loan1
decline_score <- p_CC1_Loan0 * p_Online1_Loan0 * p_Loan0
prob_E <- accept_score / (accept_score + decline_score)
prob_E
```

#I compared probability B and E.
```{r}
c(
  "B: Pivot table" = prob_B,
  "E: Naive Bayes" = prob_E
)
```

#I ran the Naive Bayes by using the training data.
```{r}
library(e1071)
nb_model <- naiveBayes(
  Loan ~ CC + Online,
  data = train,
  laplace = 0
)
nb_model
```

#The final probability for a customer with CC=1 and Online=1
```{r}
```

```
new_customer <- data.frame(
  CC = factor("1", levels = levels(train$CC)),
  Online = factor("1", levels = levels(train$Online))
)
customer_probabilities <- predict(
  nb_model,
  newdata = new_customer,
  type = "raw"
)
customer_probabilities
prob_G <- customer_probabilities[1, "1"]
c(
  "E: Manual calculation" = prob_E,
  "G: Model calculation" = prob_G
)
```